TRIASHA SARKAR

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Machine Learning Engineer with an aerospace background, building retrieval, recommendation, and scientific ML systems. I care about clear metrics, production readiness, and knowing when a model should not ship.

EDUCATION

MS, Aerospace Engineering | Georgia Institute of Technology, Atlanta, GA | Aug 2026

CGPA: 3.7/4.0 | Entered PhD program; transitioned to MS and completed the degree in Aug 2026

MSc, Aerospace Vehicle Design | Cranfield University, United Kingdom | Feb 2023

CGPA: 3.6/4.0 | First Class Honours | Thesis on geometry optimization under FAA constraints

B.Tech, Aerospace Engineering | SRM Institute of Science and Technology, India | Jun 2021

CGPA: 3.8/4.0 | First Class Honours

TECHNICAL SKILLS

ML and GenAI: PyTorch, scikit-learn, Transformers, PEFT/LoRA, RAG, BM25, dense retrieval, reranking, recommender systems, uncertainty estimation

Systems: Python, Go, SQL, Kafka, FastAPI, Docker, Kubernetes, GitHub Actions, PostgreSQL/pgvector, Prometheus, OpenTelemetry, ONNX, Core ML, Qualcomm QNN

Scientific computing: NumPy, SciPy, pandas, NetworkX, OSMnx, statistical inference, design of experiments, surrogate modeling

PROFESSIONAL EXPERIENCE

Graduate Research Assistant under Prof. Dimitri Mavris | Georgia Tech ASDL | May 2025 - Aug 2026

- Built and tested Python models for projects in simulation, airline safety, and sustainable aviation; shared reproducible analyses with faculty and sponsors.
- Reviewed assumptions and limits with aerospace experts and documented each workflow so other researchers could rerun it.

Machine Learning Engineer | Rolls-Royce | Jul 2023 - Apr 2025

- Developed Python pipelines to flag unusual engine behavior, support maintenance planning, and combine sensor streams sampled at different rates.
- Mapped certification requirements to data and model checks, then reviewed results with lifecycle engineers before use.

Data Science Intern | Rolls-Royce DataLabs | May 2021 - Jul 2021

- Cleaned aircraft-engine sensor data, built features, and compared prediction and anomaly models; summarized recurring failure patterns for engineers.

TECHNICAL PROJECTS

AIRFAANS | Geometry-aware CFD surrogates | Aug 2026

- Benchmarked three neural-network architectures for airflow prediction across three matched runs and 200 official AirfRANS test meshes per model. No architecture led every flow and force metric.

AeroRAG-X | Retrieval, post-training, and ML systems | Oct 2025 - Present

- Built a search and RAG system over 3,233 NASA report sections using keyword search, embeddings, reranking, evidence checks, and source-linked citations.
- Evaluated a frozen search baseline on 888 questions with human-selected evidence; relevant evidence appeared in the top 10 results 76.24% of the time. Answer quality was not part of this test.

IntegrityBench | Moderation evaluation and release controls | Aug 2026 - Present

- Trained a content-moderation model on Civil Comments, selected thresholds on validation data, and evaluated the frozen model on 2,802 human-labeled ToxicChat prompts.
- False acceptance rose from 1.84% to 59.32% on ToxicChat. The release stayed blocked, and the failed transfer result remains reported.

NewsLens | Recommendation and real-time search | Jul 2026 - Aug 2026

- Used chronological splits to keep future clicks out of training. The recommender scored 0.366 NDCG@10, but the uncertainty interval still included no improvement.
- Built a Go, Kafka, PostgreSQL, and FastAPI ingestion path for new articles. In a 500-event run, publish p99 was 44 ms and search freshness p95 was 79 ms.

EdgeGenBench | Real-flight ML and on-device inference | Aug 2026 - Present

- Trained a flight-anomaly model on NASA DASHlink recordings and tested it on 17,780 approaches from held-out aircraft. Its 0.738 macro F1 missed the release target.
- Exported the model to ONNX and tested corrupted sensor inputs; prediction agreement stayed above 99.55% across the tested cases.

AeroSynth-Eval | Multimodal AI evaluation | Aug 2026 - Present

- Tested generated-image augmentation for aircraft-defect classification. Macro F1 improved from 0.388 to 0.442, but crack recall fell from 40.00% to 31.67%, so the change was rejected.
- Prepared 1,735 public human preference votes for a general image-quality evaluator; the labels do not validate aircraft inspection.

Surrogate Model Learning | Real-data reliability studies | Feb 2026 - Present

- Evaluated surrogate models on public airfoil and building datasets using grouped splits, so related designs could not appear in both training and test data.
- Confirmed the uncertainty method on separate UCI concrete data. Test R^2 was 0.902, and normalized 90% intervals covered 95.83% of held-out mixture groups.

Atlanta Mobility Resilience Digital Twin | Jun 2026 - Present

- Built an Atlanta road-closure simulator using OpenStreetMap and 50 Census tract origins representing an estimated 216,659 residents.
- Added 101 mapped hospitals, clinics, fire stations, and shelters, then carried ACS population uncertainty through the accessibility calculations. Observed traffic calibration is still pending.

Silent Failure Detection in Rocket-Motor Simulations | Georgia Tech ASDL | May 2026 - Jul 2026

- Traced silent failures in a 15,120-case rocket-motor study to an uncapped convergence loop; a leakage-safe classifier reached 96.8% accuracy on unseen geometries.

HERO | Source-aware retrieval for airline safety | Georgia Tech ASDL | May 2025 - Aug 2026

- Developed retrieval and evaluation methods that link AI-generated airline-safety answers to technical sources analysts can inspect.

GREEN TEA and Project EAGLE | Sustainable aviation modeling | Sep 2025 - Apr 2026

- Built surrogate models to speed up sustainability calculations and connected fuel, demand, transport, and life-cycle inputs; GREEN TEA remains in sponsor use.

Equity Backtest | Walk-forward signal evaluation | Jul 2026

- Backtested a stock-momentum signal using only data available at each date and included trading costs. The signal disappeared after costs and missed the pre-set significance threshold.

LEADERSHIP

Graduate Chair | Women of Aeronautics and Astronautics, Georgia Tech | Aug 2025 - Aug 2026